

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**First Semester of B. Tech (EC/IT/CE) Examination****December 2014****EC 101 Basics of Electronics Engineering****Date: 05.12.2014, Friday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below.****[07]**

- a. Two Electric bulbs rated for the same voltage have powers of 200W and 100W. If their resistances are R_1 and R_2 respectively, then

- | | |
|--------------------|---------------------|
| (i) $R_1 = 2 R_2$ | (iii) $R_2 = 4 R_1$ |
| (ii) $R_1 = 4 R_2$ | (iv) $R_2 = 2 R_1$ |

- b. An Electric circuit contains _____.

- | | |
|----------------------------|--|
| (i) Active Elements only | (iii) Active and Passive Elements both |
| (ii) Passive Elements only | (iv) None of the above |

- c. The Positive terminal of a 12V battery is earthed. The negative terminal will be at ____.

- | | |
|--------------------|-----------------|
| (i) Zero Potential | (iii) +12 Volts |
| (ii) -12 Volts | (iv) 24 Volts |

- d. The Capacitance of parallel plate capacitor depends upon:

- | | |
|--------------------------------|--|
| (i) Type of metal used | (iii) Thickness of plates |
| (ii) Separation between plates | (iv) Potential difference between plates |

- e. An Ammeter is _____ Instrument.

- | | |
|-------------------|------------------------|
| (i) An Indicating | (iii) An Integrating |
| (ii) A recording | (iv) None of the above |

- f. A _____ provides a visual representation of any waveform applied to the input terminals.

- | | |
|-----------------|-------------------------|
| (i) Multi-meter | (iii) CRO |
| (ii) Watt meter | (iv) None of the above. |

- g. A transformer will work on _____.

- | | |
|---------------------|-------------------------|
| (i) AC only. | (iii) DC only |
| (ii) AC and DC both | (iv) None of the above. |

Q - 2.a Derive the equations of E.M.F. for transformer.

[04]

Q - 2.b Answer any two questions.

[10]

- (i) Explain input and output characteristics of CE configuration, also derive relationship between α and β .
- (ii) Convert $(57)_5$ in decimal and binary and hexadecimal.
- (iii) Explain AND & NOR Gate. Find 1's complement of $(100101)_2$.

Q - 3 Answer any one.

[14]

- a.
 - 1. Explain Delta- star and Star- Delta Transformation
 - 2. Define Electric Field, Electric flux Density and Electric field Intensity and derive the relationship between them.
- b. With neat and clean diagrams, explain Full-wave bridge rectifier and derive its equation for efficiency as well as Ripple factor.

SECTION - II

Q - 4 Answer the question below.

[07]

- a. A transformer does not possess _____ changing property.
 - (i) Impedance
 - (ii) Voltage
 - (iii) Current
 - (iv) Power
- b. The doping level in a zener diode is _____ that of a crystal diode.
 - (i) More than
 - (ii) Less than
 - (iii) the same as
 - (iv) None of the above
- c. The reverse saturation current in a diode is of the order of _____.
 - (i) mA
 - (ii) μ A
 - (iii) A
 - (iv) kA
- d. In a transistor, $I_c = 100\text{mA}$ and $I_E = 100.5\text{mA}$. The value of β is _____.
 - (i) 100
 - (ii) Nearly 1
 - (iii) 50
 - (iv) 200
- e. In a transistor, $\beta=100$ and collector current is 10mA . Then I_E is _____.
 - (i) 10mA
 - (ii) 10.1mA
 - (iii) 11mA
 - (iv) 9.9mA
- f. Make Logic gate diagram and Truth table for NAND Gate.
- g. Convert $(5C7)_{16}$ to Decimal.

Q – 5.a Convert $(125.75)_{10}$ into

[05]

- 1) Binary 2) Octal
3) Hexadecimal 4) Base 3 and 5) Base 4

OR

Q – 5.a Explain charging of a capacitor and derive the equation for charging voltage across a capacitor.

[05]

Q – 5.b Explain parallel plate capacitor having uniform dielectric medium.

[04]

Q – 5.c A Crystal diode having internal resistance $r_f = 20\Omega$ is used for half wave rectification. If the applied voltage $v = 50\sin\omega t$ and load resistance $R_L = 800\Omega$, find:

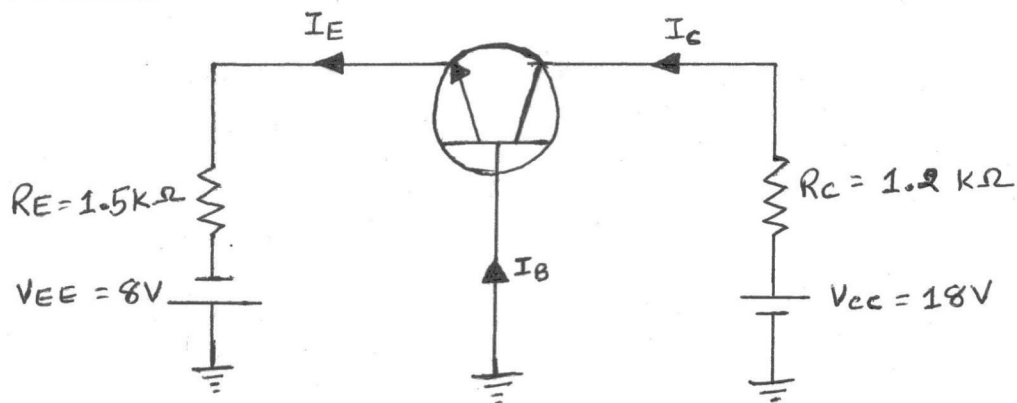
[05]

- a) I_m c) I_{rms}
b) I_{dc} d) Efficiency of rectification

OR

Q – 5.c For common base circuit shown in Figure 1. Determine I_C and V_{CB} . Assume transistor to be of silicon.

[05]



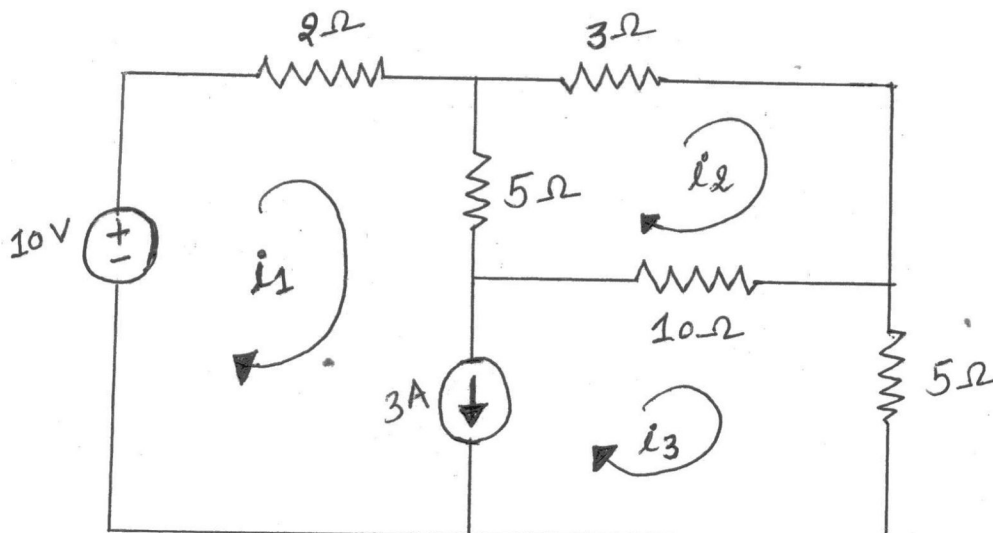
[Figure 1]

Q – 6.a Differentiate between Intrinsic and Extrinsic Semiconductors.

[02]

Q – 6.b Find the values of i_1 , i_2 and i_3 for the Figure 2.

[06]



[Figure 2]